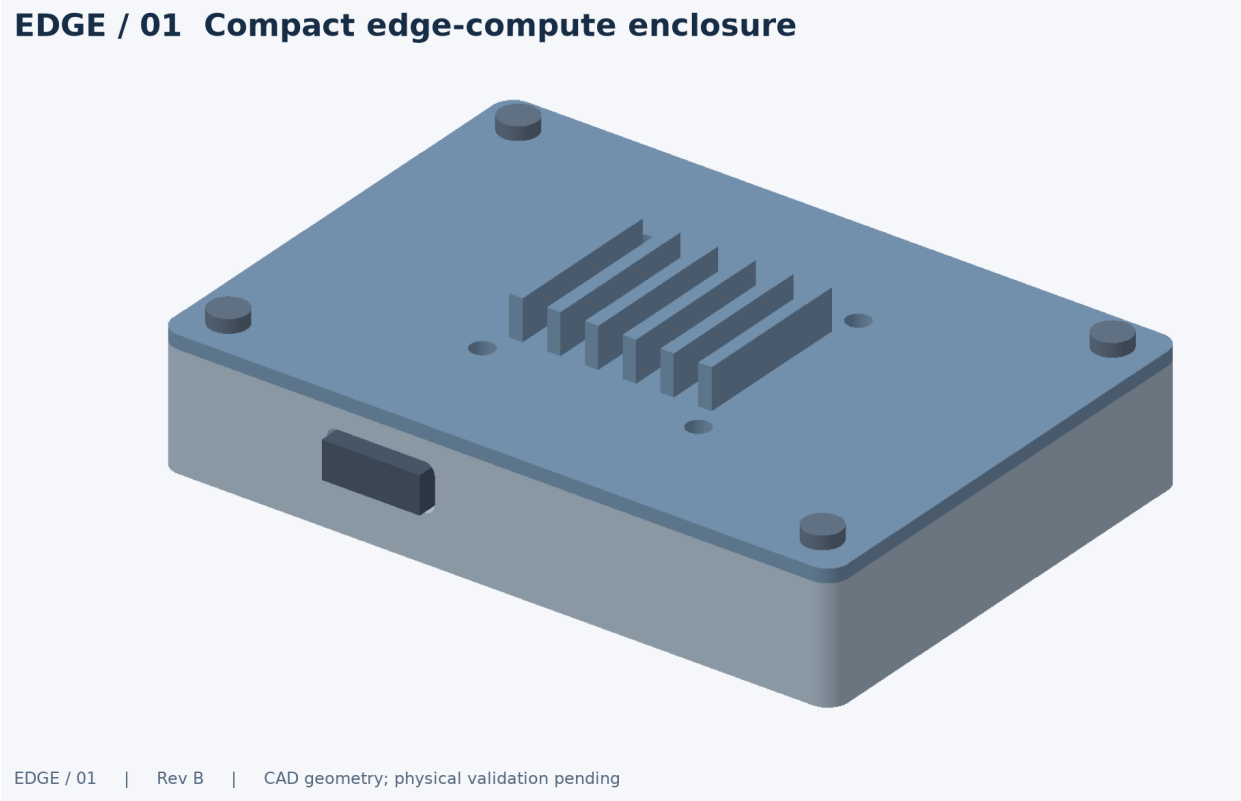
DESIGN / MANUFACTURE / VERIFY

Compact edge-compute enclosure

Parametric mechanical design, manufacturing planning and rapid validation



Envelope	Thermal path	Evidence
100 × 70 × 25 mm housing	Chip / pad / integral Al lid / air	PASS nominal digital checks

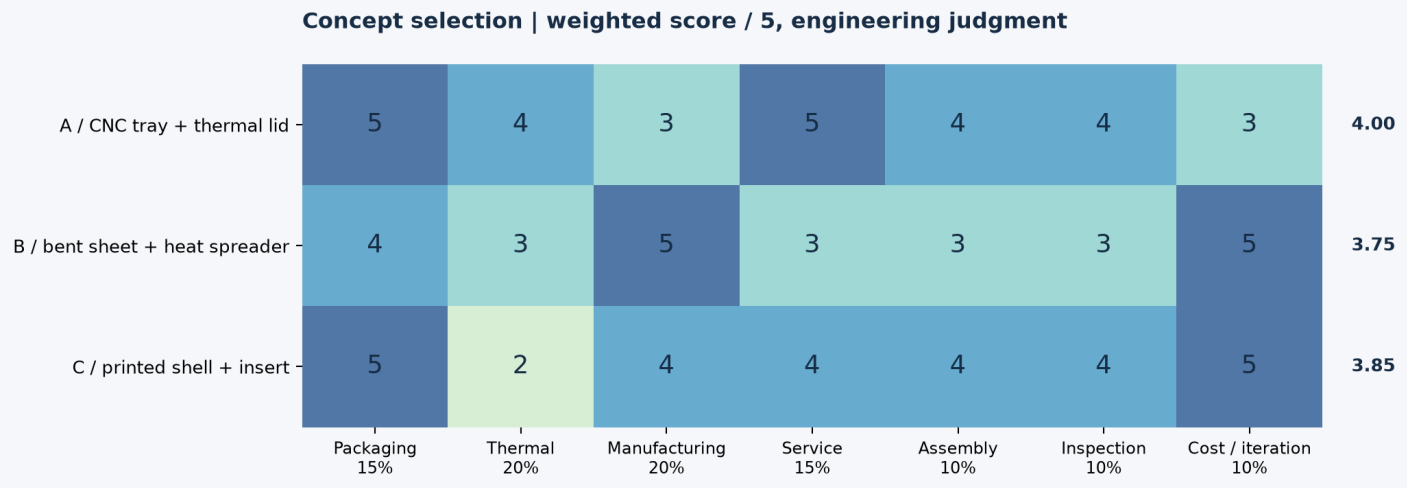
A reviewable path from ambiguous packaging requirements to editable solids, machining setups and measurable first-article tests. The useful result is not a flawless prediction: passive cooling misses the screening target, and pad compression remains a design risk.

Independent demonstrator. Not a comma.ai design or replica. No physical prototype, machining, measured thermal performance, CFD or FEA is claimed.

01 / REQUIREMENTS & TRADEOFFS

Choose the architecture deliberately.

The board and its interfaces are assumed. The target is a compact bench demonstrator with inspectable datums, removable electronics and a continuous metal thermal path. The fan variant deliberately trades height for a testable cooling hypothesis.



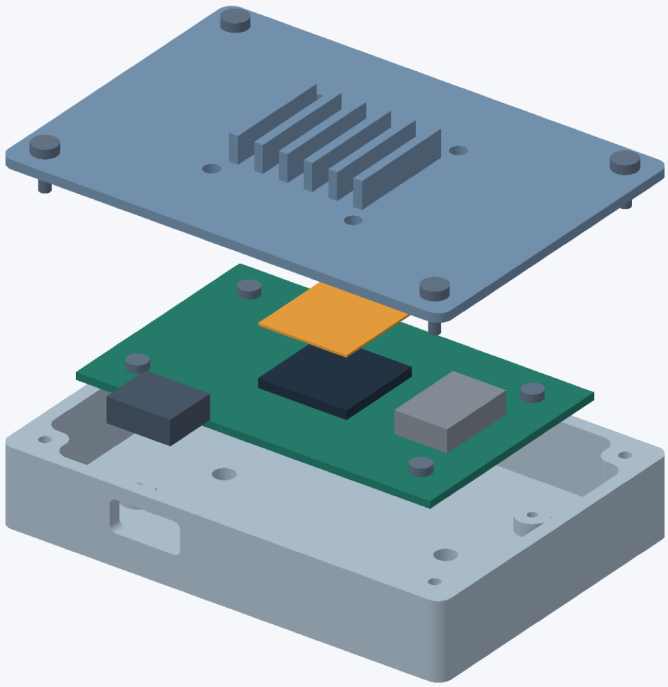
Requirement	Decision / verification
Compactness	Passive housing 100 × 70 × 25; connector adds 2 mm. Fan body to 35 mm; reserve hardware to 38 mm.
Serviceability	Four M3 lid screws; four M2.5 PCB screws. Board removal exposes external mount bolts.
Manufacturability	R3 main cavity transitions, 3.6 mm fin channels; two-sided lid nest and three tray setups.
Thermal contact	0.50 mm nominal gap. Critical Z dimensions inspected before pad selection.

Concept A scores 4.00/5 versus 3.75 for bent sheet and 3.85 for a printed shell. The narrow margin is intentional: printing remains the first fit-validation step. Scores are subjective screening, not empirical optimization.

02 / PACKAGING

Separate what must fit from what must cool.

Built for access. Designed around the thermal path.



01 Finned thermal lid
Integral pedestal

02 Thermal pad
03 Generic PCB + chip

04 Machined tray
Four PCB supports

EDGE / 01 | Rev B | CAD geometry; physical validation pending

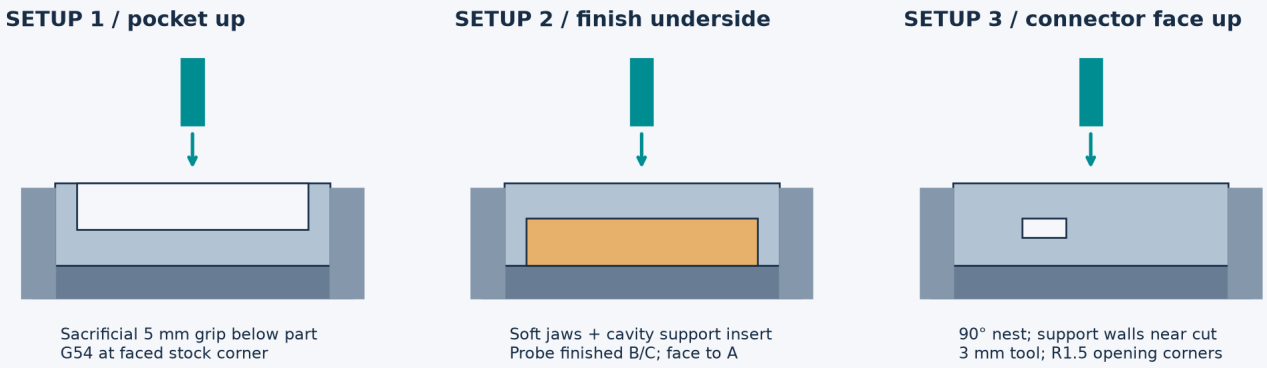
Layer / interface	Nominal geometry
Tray	3 mm floor/wall; 17 mm rim; corner islands and integral PCB supports
PCB and keepouts	80 × 50 × 1.6 at Z6; generic 20 × 20 × 2 chip; separate connector/component envelopes
Thermal lid	2 mm plate at Z17..19; pedestal to Z10.1; six fins to Z25
Hardware	Simplified M3 × 8 lid / M2.5 × 5 PCB screws; actual threads, fan wiring and nuts not fully modeled

The integral pedestal removes loose-heatsink retention ambiguity. Removing the lid separates the thermal interface, so every service cycle requires inspecting and, where the supplier requires it, replacing the pad. All PCB dimensions must be replaced with measured interfaces before ordering metal.

03 / MANUFACTURING

A process plan an engineer can challenge.

CNC setup concepts | review before programming



Illustrative setup geometry. No generated or verified machine toolpaths; clamp and tool envelopes require CAM review.

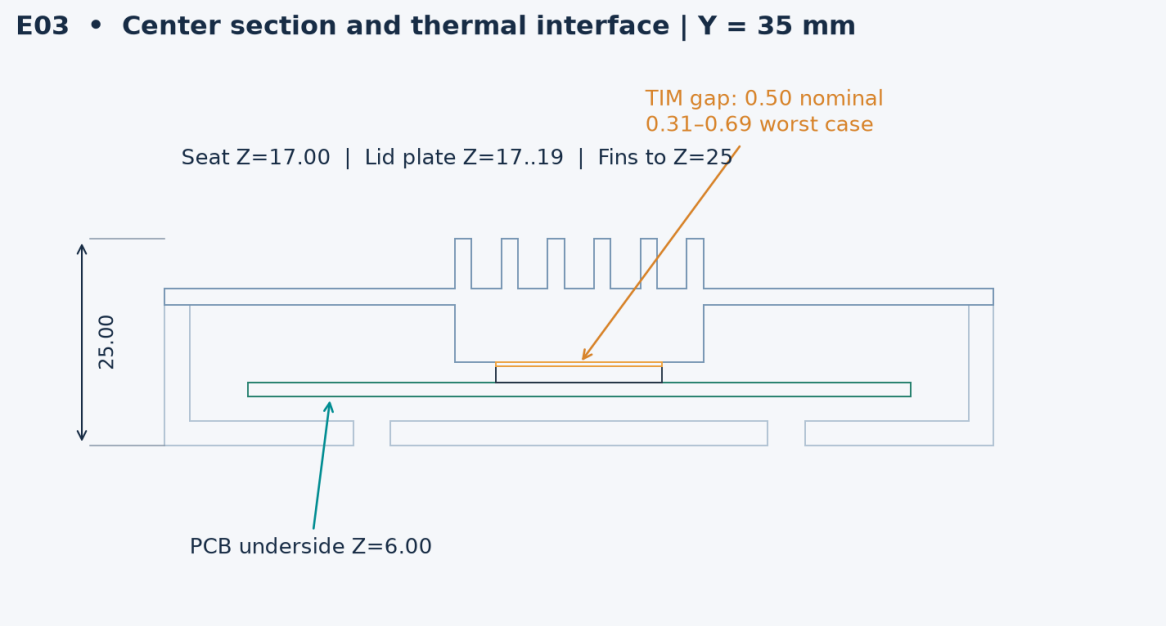
Workpiece	Stock / operation / hold
Tray	110 × 80 × 25 candidate blank. Grip sacrificial foot, pocket from top, flip into supported soft jaws, then rotate for connector opening.
Thermal lid	106 × 76 × 20 blank. Machine pedestal side, flip into relieved support nest, finish 2 mm plate and 6 mm fins. Protect against bow.
Tools	Candidate Ø6 rougher, Ø3 fin/connector finisher, spot/drill/tap tools. Verify reach and holder geometry in actual CAM.
Print precursor	PETG fit print, underside down. Front opening may need bridge support. Printed lid is an unpowered dummy, not a heat sink.

Haas VF-2 is the requested reference machine. The actual configuration, postprocessor, offsets, tool assemblies, clamps and safe simulation remain open. No G-code or production-ready toolpaths are supplied.

Rev B replaced cylindrical corner-boss junctions with a rounded stepped cavity and moved support holes inward. This improves nominal tool access; full swept-tool/holder collision verification is still required.

04 / INSPECTION & STACK-UP

Control the contact, not just the outline.



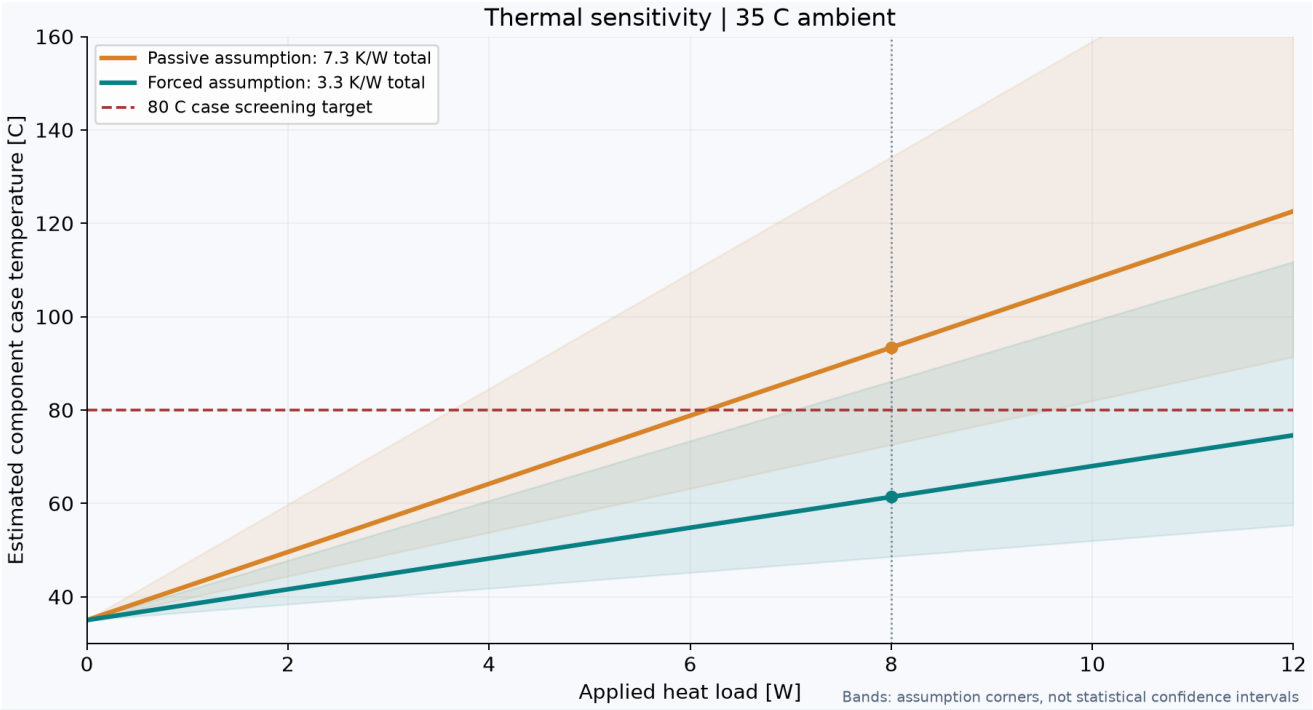
Stack term	Nominal [mm]	Tolerance [mm]
Base rim from A	17.00	±0.03
Minus pedestal projection	6.90	±0.03
Minus PCB support height	6.00	±0.03
Minus PCB thickness / chip height	1.60 / 2.00	±0.05 / ±0.05
Resulting gap	0.50	Worst case 0.31..0.69

A free pad 1.00±0.05 mm would see 27.4..70.5% compression. This is not a released specification: select a supplier pressure curve, measure the real stack and revise the pedestal or shims before loading the board.

A/B/C locate the base. Proposed support height ±0.03 and flatness 0.05 control seating. Hole position uses an actual cylindrical zone and CMM/qualified mandrel inspection; caliper coordinates alone do not certify it. Full dimensions and acceptance methods are in the drawing package.

05 / THERMAL ENGINEERING

The passive hypothesis fails the screen.



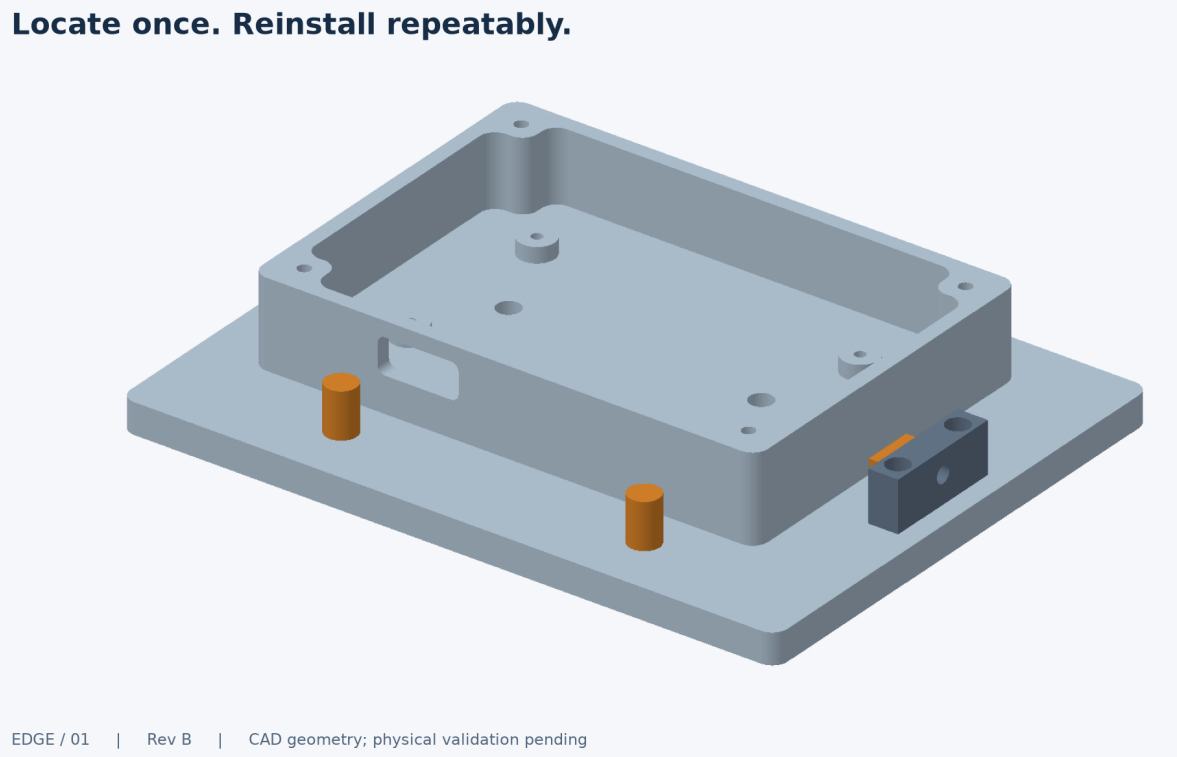
At 8 W /35°C ambient	Passive	Forced hypothesis
Total thermal resistance	7.3 K/W	3.3 K/W
Predicted case temperature	93.4°C / exceeds 80°C	61.4°C / nominally below 80°C
Uncertainty corner range	72.6..134.2°C	48.6..86.2°C

ΔT = Q × Rθ. The network assumes 0.8 K/W pad/contact,0.5 K/W spreading, and 6.0 or 2.0 K/W effective air resistance. It estimates case temperature, not junction temperature. Bands are assumption corners, not confidence intervals.

Measure heater-terminal voltage/current and case/lid/ambient temperatures. Start at 2 W, increase only with margin, and stop at case 80°C, accessible metal60°C, sensor loss, fan failure or any lower component limit. Independent cutoff and current limiting are required before energizing.

06 / FIXTURE & VALIDATION

Repeatable location needs a repeatable procedure.



Constraint	Implementation
Z / roll / pitch	Three noncollinear support buttons on datum A
Y / yaw	Two front side buttons touching datum B
X	One left button touching datum C
Bias / retention	Two gentle lateral screw pads near the floor; gravity seating. Not a machining clamp.

Use ten complete remove-and-reinstall trials per operator, two operators, with indicators left at the original zero. Proposed per-operator and pooled range ≤0.05 mm; between-operator mean shift ≤0.03 mm. Raw templates intentionally contain no fabricated data.

Replaceable locator buttons can be shimmed; block mounting bores and lateral screw pilots are modeled. Selected screws, nuts, bench feet and a torque-resisting hold-down remain completion items. Do not use this inspection jig for cutting forces.

07 / RELEASE STATUS

Evidence today. Physical proof next.

31 component export pairs checked; 31 tests executed, 0 failures; 361 assembly pair checks.

Evidence class	What this package establishes
Generated and checked	Editable CadQuery B-reps; component STEP round trips; watertight STL; passive/fan assembly solid counts; nominal intersections.
Calculated	Thermal resistance sensitivity, worst-case pad gap, concept weights and geometric clearances.
Planned	CNC setups, tool candidates, inspection methods, printing trials and fixture/heater protocols.
Not established	Real-board compatibility, pad force, structural strength, thread durability, actual thermal performance or fixture repeatability.

Next physical work

1. Measure/select electronics, pad, fan and fasteners; review blind-hole depths.
2. Print an unpowered fit article and hole/bridge coupon; inspect and iterate.
3. Build the inspection fixture and run 20 reinstall trials.
4. Simulate machine-specific CAM, machine the metal parts and inspect critical heights.
5. Establish pad contact and run guarded heater tests with raw logs.

Three largest risks

Uncertain effective cooling; pad pressure/PCB strain across the tolerance stack; thin-part distortion and tool access during actual machining.

Sources and reproducibility

Public context: comma four announcement, commaai/hardware, comma jobs. No source geometry imported. Full source notes: report/sources.md. Run **python scripts/rebuild.py** from the repository root after installing requirements.txt.

github.com/gokulg846/compact-edge-enclosure